

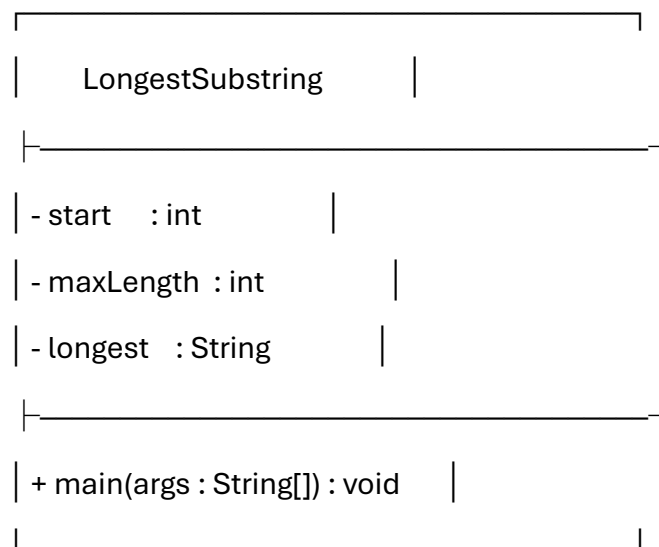
Aim

To write and execute a Java program to find the **longest substring without repeating characters** in a given string and display its **length and substring**.

Algorithm

1. Start the program.
2. Read a string from the user.
3. Initialize variables to store the starting position, maximum length, and longest substring.
4. Use two pointers to examine the characters of the string.
5. Check whether the current character is already present in the current substring.
6. If it is repeated, move the starting position forward.
7. Continue until all characters are checked.
8. Keep track of the longest substring found.
9. Display the longest substring and its length.
10. Stop the program.

UML Diagram:



Code:

```
package Records;
```

```
import java.util.Scanner;

public class Strings {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        int start = 0;
        int maxLength = 0;
        String longest = "";

        for (int end = 0; end < str.length(); end++) {

            char current = str.charAt(end);

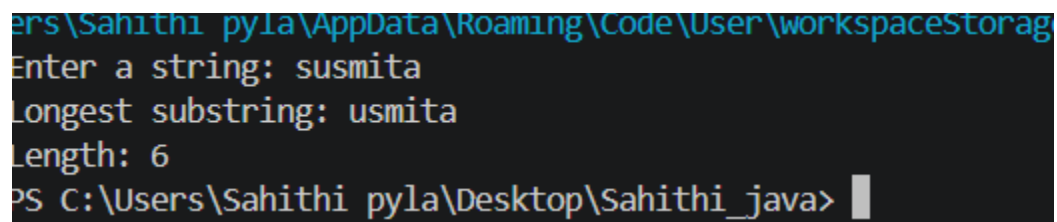
            // Check for repeated character
            int index = str.indexOf(current, start);

            if (index != end) {
                start = index + 1;
            }

            int currentLength = end - start + 1;
```

```
        if (currentLength > maxLength) {  
            maxLength = currentLength;  
            longest = str.substring(start, end + 1);  
        }  
    }  
  
    System.out.println("Longest substring: " + longest);  
    System.out.println("Length: " + maxLength);  
  
    sc.close();  
}  
}
```

Output:



```
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java> Enter a string: susmita  
Longest substring: usmita  
Length: 6  
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>
```